Nicholas Xydis

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Education

Concordia University — Expected 2029
Certificate of Science and Technology (Incoming Bachelor of Computer Science) — *Montreal, QC*

Experience

IT Administrator & Operations — April 2022 - February 2026
Centre D'Auto Allard — *Montreal, QC*

- Developed the company website using TypeScript and React, serving 50+ daily users and contributing to a 55% increase in customer appointments.
- Implemented a SQL database for inventory and operational data, reducing manual tracking by 7 hours per week.
- Managed Google Business Profile and analytics, generating 40+ reviews and 1,200+ monthly profile views.
- Coordinated customers, parts suppliers, and service vendors across procurement and scheduling.
- Maintained computers, networking equipment, and security systems, resolving hardware and connectivity issues.

Projects

BankLite | *C#, PostgreSQL, Docker, Linux, JavaScript* — March 2026 - September 2026

- Independently engineered a full-stack banking application with secure authentication, real-time peer-to-peer payments, an AI financial assistant, and bilingual English and French support, reaching 1,400+ unique visitors.
- Load tested with k6, processing 533,000+ requests at 1,184 req/s and 0.00% failures under 1,000+ concurrent users.
- Hardened a Linux VPS with Nginx, Cloudflare, CI/CD, 24/7 monitoring, and seven custom rate limits.
- Validated reliability and security with ACID transactions, 482 tests (448 backend, 34 E2E), and OWASP ZAP.

Stage | *Python, SQLite, Textual* — July 2026 - September 2026

- Engineered an internship aggregation platform spanning 1,450+ employer boards across 15 ATS platforms, centralizing relevant CS roles into one searchable database, and surpassing 500 PyPI installs since launch.
- Built a one-command, zero-config CLI & TUI handling 146,000+ postings per sync with full-text search.
- Automated new-role delivery via Discord webhooks and scheduled syncs across macOS, Linux, and Windows.
- Verified reliability with 2,362 automated tests at 89% coverage across rate limits, circuit breakers, and retry backoff.

CS Resource Hub | *Python, TypeScript, React, Vite* — June 2026 - September 2026

- Architected a Python-powered data platform helping Canadian Computer Science students discover 330+ curated resources, with a single JSON dataset deterministically generating the website and project artifacts.
- Created a contribution pipeline enforcing schema, duplicate, and link checks pre-merge while generating 6 artifacts.
- Orchestrated CI/CD across 15 GitHub Actions workflows with approval-gated Cloudflare Pages deployments.
- Executed 375 tests, 5 browser targets, and 22 viewports, plus CodeQL and 100/100 Lighthouse scores.

Extracurriculars & Certifications

International Finalist — September 2026 - Present
McGill-FIAM Asset Management Hackathon — *Montreal, QC*

- Designed a LightGBM market neutral equity strategy, backtesting 12.8% net annualized returns over 5.5 years.
- Deployed 7 AI agents acting as rival investors across 265K trades, catching hidden risk in 42% of our stocks.

Quantitative Analyst — September 2026 - Present
Quantitative Research and Competitions Club (QUARCC) — *Montreal, QC*

- Developing HeapStreet, a C++ quant game where players outcode 12 AI rivals in 470+ algorithmic challenges.
- Contribute to weekly research meetings, helping organize club events for the quant finance community.

Options 101 | *Akuna Capital* — September 2026

Junior Programmer Pathway | *Unity* — March 2026

Technical Skills

Languages: Python, C#, C++, SQL, TypeScript, JavaScript
Data & ML: PyTorch, scikit-learn, Pandas, NumPy, XGBoost, LightGBM, Matplotlib, Jupyter
Frameworks & Databases: .NET, ASP.NET Core, React, Next.js, Tailwind CSS, PostgreSQL, SQL Server, SQLite
DevOps & Tools: Linux, Docker, Git, Bash, Azure, GitHub Actions, PowerShell, Nginx, Playwright, Unity
Cybersecurity: Nmap, Wireshark, Burp Suite, Metasploit, OWASP ZAP
Spoken Languages: English & French (Bilingual)